

Filip Szalewicz

Principal Engineer / Head of Engineering | Agentic AI, Digital Twins, Multi-Agent Systems

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Summary

Principal engineer and Head of Engineering with 20+ years of experience building SaaS platforms, data systems, AI products, and cloud infrastructure. Hands-on technical contributor and people leader who drives impact across the enterprise: designing digital twins, multi-agent systems, and Agentic OS patterns that help organizations adopt AI directly inside active delivery workflows. Partner with executive leadership, technical leaders, and engineering teams to train AI best practices, implement governance and human-in-the-loop controls, and accelerate measurable outcomes across product and operations. Recent work includes organizational transformation for a 150-person technology company; agent-powered SaaS products such as vizly.ai v2, LookingGlass Lifestyle v2 (web and iOS), and Data Reaktor that use AI LLMs for data analysis, image processing, enrichment, and classification; and acting as an agent of change driving AI-first enablement and adoption across client organizations.

Core Competencies

Agentic AI Strategy & Transformation - AI adoption strategy, operating models, organizational change, delivery acceleration

Digital Twins & Multi-Agent Systems - twin design, agent factories, agent orchestration, context design, workflow decomposition

Agentic OS & Workflow Automation - ecommerce and business workflow automation, human-in-the-loop approvals, guardrails, self-improvement loops

Agent-Powered SaaS & Data Products - AI-assisted analytics, image processing, lead intelligence, big-data enrichment and classification

AI Governance & Evaluation - approval flows, auditability, guardrails, feedback loops, measurable outcomes

Cloud Runtime Engineering - Python, FastAPI, Node.js, Next.js, Supabase, Docker, Kubernetes, Cloud Run, AWS, GCP, Azure

Data & ML Platforms - ETL/ELT, Airflow, Spark, BigQuery, SageMaker, data lakes, warehouses, reporting, analytics

Leadership - hands-on engineering, people management, mentoring, executive partnership, enterprise delivery impact

Professional Experience

Solid Cage Inc Macomb, MI

Principal Engineer / Head of Engineering, Agentic

AI 2024 - Present

- Lead agentic-first transformation programs for multiple clients, including a 150-person technology company, partnering with executive leadership, technical leaders, and engineering staff to accelerate AI adoption across the organization.
- Contribute as a principal engineer while leading people and delivery, shipping production systems and coaching teams so AI lands inside live workstreams rather than side projects.
- Train leaders and delivery teams on practical AI best practices, workflow redesign, and operating models that embed AI directly into projects already in flight instead of isolated experiments.
- Drive development and adoption of digital twins in preparation for agent factory deployment, defining roles, context, guardrails, and governance patterns for production use.
- Architect Agentic OS patterns for end-to-end ecommerce workflow automation using human-in-the-loop approvals, guardrails, structured context, and self-improvement loops.

- Act as an agent of change driving AI-first enablement and adoption across client organizations, helping teams embed AI into live delivery so throughput and product outcomes improve.
- Built and open-sourced reusable agentic infrastructure and adoption assets, including agent-kernel, agent-factory, agentic-playbook, and operational-intelligence-lab.
- Built and advised on agent-powered SaaS products including vizly.ai v2, LookingGlass Lifestyle v2 (web and iOS applications), and Data Reaktor, using AI LLMs for data analysis, image processing, big-data enrichment and classification, and related production workloads.

Solid Cage Inc Macomb, MI

Principal Consultant & SaaS Founder 2009 - 2023

- Founded and scaled consultancy delivering 50+ SaaS, data, and analytics solutions with 95%+ client retention.
- Architected PathGnos for Mayo Clinic, an Azure-based digital pathology archive supporting DICOM/SVS formats, Cosmos DB storage, CI/CD, and high availability.
- Designed and deployed web applications, data pipelines, data marts, dashboards, and reporting systems for SMB and enterprise clients.
- Advised teams on cloud adoption, engineering execution, ETL practices, and secure architecture across healthcare, fintech, and SaaS environments.

Shoptelligence Ann Arbor, MI

Vice President of Engineering 2022 - 2025

- Architected a 15+ TB data lake and warehouse platform, reducing analytics retrieval from 3+ weeks to 1 day.
- Built ETL pipelines with Airflow, Spark, Data Fusion, and BigQuery, boosting integration speed by 75% and supporting \$1M+ ARR growth.
- Established data quality standards and CI/CD practices that reduced support tickets by 500%.
- Expanded an omni-channel retail SaaS platform with AI-assisted personalization and modern data architecture.

West Monroe Partners Chicago, IL

Senior Architect 2021 - 2022

- Delivered data strategy, warehouse design, and analytics modernization programs for private equity clients.
- Trained consultants in SQL, Python, and data architecture practices.

Rocket Mortgage (Quicken Loans) Detroit, MI

Principal Engineer / Team Lead, Data Science 2017 - 2021

- Architected a petabyte-scale data lake and ELT platform using AWS Glue, Spark, and Cloudera, reducing annual cost by more than \$1M.
- Led enterprise data fabric and MDM integrations that improved compliance and increased loan volume by 10%+.
- Operationalized ML platforms on AWS using SageMaker, Lambda, and Athena for fraud detection and risk modeling.
- Mentored 50+ engineers on ETL, Spark, AWS, and CI/CD; recognized with the "Automate All the Things" award in 2020.

360ofme Inc Royal Oak, MI

CTO / Principal Engineer 2016 - 2017

- Designed a high-volume SaaS platform for secure personal data management using Node.js, MongoDB, PostgreSQL, and RabbitMQ.
- Established CI/CD automation and real-time monitoring across AWS and Heroku environments.

Earlier Roles

1993 - 2016

- Healthmaster Holdings - Built SaaS medical software for K-12 schools and EDI medical billing systems using 837/835/997 healthcare transaction flows.
- imX Solutions - Lead developer for a patented secure digital payments and authentication platform.
- LogicaCMG / JDM Systems / New Wave Software - Delivered reporting, integration, ERP, and plant-floor systems for enterprise and industrial clients.

Selected Public Agentic AI Work

agent-kernel

Reusable digital twin knowledge layer with agent skills, prompts, templates, diagrams, and automation for consistent execution.

agent-factory

FastAPI runtime for hosting digital twins with model routing, Supabase-backed persistence, admin governance, and Cloud Run deployment patterns.

agentic-playbook

Role-based adoption layer for engineers, architects, executives, compliance leaders, and power users implementing agentic AI.

operational-intelligence-lab

Open methodology connecting AI implementation to rate-of-improvement, ROI modeling, training, and business deployment.

Technical Skills

- Languages & Frameworks: Python, SQL, Java, Node.js, Next.js, TypeScript, JavaScript, C#, FastAPI
- Agentic / AI: AI agents, digital twins, multi-agent systems, Agentic OS, OpenAI, Anthropic, xAI/ Grok, prompt design, context engineering, HITL, guardrails, evaluation loops
- Cloud Platforms: AWS (EC2, Glue, S3, Lambda, SageMaker, Athena, QuickSight), GCP (BigQuery, Data Fusion, Vertex AI, Airflow, Looker, Kubernetes, Cloud Run), Azure (Cosmos DB, Web Apps, Functions)
- Data Technologies: Data lakes, data warehousing, ETL/ELT, Airflow, Data Fusion, Spark, Hadoop, Hive, PostgreSQL, MongoDB, DynamoDB, SQL Server, MySQL
- DevOps & Tools: Git, Jenkins, GitLab, Docker, Kubernetes, Terraform, Maven, CircleCI, Azure DevOps, n8n, Supabase, TDD, Agile/Scrum
- Analytics & BI: Tableau, Looker Studio, Power BI, semantic layer design, KPI dashboards
- Compliance: HIPAA, CCPA, GDPR-aligned data practices, EDI, DICOM

Patents, Certifications, and Education

- Patents: System, Method, and Apparatus for Conducting Secure Online Monetary Transactions; Optical Disc Design Patent
- Certifications: AWS Certified Developer - Associate (2018); AWS Certified Solutions Architect - Associate (2018); Microsoft Certified Professional: Requirements & Solution Architecture (2000)
- Education: Oakland Community College - Associate's in Computer Information Systems; University of Phoenix - Information Systems Management coursework; Ongoing professional development in cloud, ML, data engineering, and agentic AI systems